



Devesh Kumar
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Microelectronics

16D070044
UG Third Year (Dual Degree)
Male
DOB: 11/02/1997

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2019	8.28
Intermediate/+2	CBSE	St Joseph Public School, Kota	2015	84.00
Matriculation	CBSE	Delhi Public School, Patna	2013	9.60

SCHOLASTIC ACHIEVEMENTS

- Pursuing minor degree in Computer Science and Engineering Department
- Was in top 1% in JEE advanced 2016
- Secured 99.1 percentile in JEE mains 2016
- Qualified for National Talent Search Exam(NTSE)
- Received certificate of high distinction in Australian National Chemistry Quiz
- Qualified for first stage of National Mathematics Talent Contest, 2014
- Recipient of IIT Bombay's MCM Scholarship based on my performance

PROJECT TAKEN

➤ Face Recognition

Course project under Ajit Rajwade Computer Science Department (CS663- Fundamentals of image processing)

- Developed a Facial recognition program on Matlab based on Fisher Discriminant Analysis and Principle component analysis and did a comparison between them
- The program is capable of handling lighting and facial variation between the dataset and the input.

➤ Text summarization

Course project under Sunita Sarawagi Computer Science Department (CS419- Introduction to Machine learning)

- Made a Recurrent Neural network for natural processing task that could take along sentence as an input and output a summarized version of it. Coded on Python and used keras and tensor flow libraries

➤ RISC architecture Implementation

Course project under Virendra Singh Electrical Engineering Department (EE309- Microprocessors)

- Designed a RISC based multicycle 16bit processor using point to point communication infrastructure.
- Coded a total of 14 instructions with three machine code instruction formats on VHDL and then demonstrated on a FPGA board. Did pipelining to reduce the CPI of the program

➤ Wireless Encrypted Messenger

Summer project under Madhav P.Desai, Electrical Engineering Department

- Developed a messaging system to wirelessly exchange encrypted messages between two complex programmable logic devices (CPLDs) and display them on a LCD in real time.
- Wrote the code from scratch and optimized it to make it more efficient and reliable
- Configured the keypad to take 32 key input including A-Z, backspace, space, enter and delete.

➤ Reaction Time Calculator

Course project under Madhav P. Desai Electrical Engineering Department (EE214-Digital Circuit Lab)

- Programmed a CPLD to calculate the reaction time of a user between a LED glowing and pressing of a react button and displaying the total score after five rounds on a LCD in real time.
- Made it foolproof by disqualifying the user if he/she presses the button before the LED glows

➤ Universal Signal Processing

Course project under Siddharth Tallur, Electrical Engineering Department (EE230- Analog lab)

- Designed a flexible analog signal circuit capable of being reconfigured according to the need
- The circuit converts the analog any input analog signal to the input range of devices like ADC or CPLD

➤ Other CS Implementation and Projects:

- Implemented probabilistic classifier and decision tree on Python and tested it on Kraggle
- Implemented many image enhancement algorithms on Matlab from scratch including adaptive histogram equalization and corner limited adaptive histogram equalization
- Also implemented several image segmentation technique including harris corner detection and mean shift segmentation and edge detection.

INTERNSHIPS

➤ Speed Labs | Mumbai, India

[DEC,17]

Architecture Improvement for Edutech platform| Worked with software development and content team

APPROACH	Analyzed the software architecture from user perspective and recommended new features to improve overall experience for student Established chapter wise timeline & lead plan; designed & Ideated new patterns and presentations
IMPACT	Many of the recommended features were added to the platform; Improved the utility of the existing content by adding more patterns; Significantly improved the overall presentation and quality

POSITION OF RESPONSIBILITY

➤ Manager |Aavriti,18

[Jan,18-Mar,18]

- Events manager in the team of 8 members responsible for the electrical engineering department fest.
- Organized various events, like electroburst and circuitreck, with a footfall of over 500 students from different colleges all over Mumbai.

➤ Team Shunya

[Aug,17-Dec,17]

Only team to represent India in Solar Decathlon Europe, 2014|Constructed house with net positive energy req. worked in electrical division.

- **Lighting Automation:** Planned lighting of the house for solar decathlon, 2018. Sensor controlled lighting which automatically adjust according to ambient sunlight in the room.
- **Appliances:** Made a database of products to be used in the house, characterized them according to their energy needs and integrated the power consumption of all devices to get the load curve.

➤ Organizer | Entrepreneurship Cell, IIT Bombay

[Jan,17- Mar,17]

A non profit organization that aims at manifesting the latent entrepreneurship spirit

- Active team member of the Cooperative Relation (CR) group
- Organized, planned and executed various events in E-summit 2017, like Eureka

TECHNICAL SKILLS

Languages	C, C++, Python, Matlab, VHDL
Platforms	Windows, Mac, Ubuntu
Tools and Packages	Solidworks, Autodesk, Ngspice, Latex, Xcircuit, GNUPlot, Keil

EXTRA CURRICULAR ACTIVITIES

- Green Campus NSS:
 - Actively contributed in National Social Service group for conservation of flora at campus.
 - Made placard for proper identification of trees in the campus.
 - Made motivational videos for people in their local language to start gardening.
- Participated in VIII IIT Bombay national debate competition.